

Course information

Radoslav Neychev
Vladislav Goncharenko



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Outline

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2. How to learn?
3. Prerequisites
4. Communication
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6. Grading

Team

**girafe
ai**

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Radoslav Neychev

- Founder of girafe-ai
- Author of machine learning courses at MIPT, MSU, and YSDA
- author of the Master's program "Modern Methods of Artificial Intelligence"
- Participated in the implementation of AI in experiments at the Large Hadron Collider.
- Areas of scientific interest: Artificial Intelligence; Large Language Models, Reinforcement Learning, Human Activity Recognition



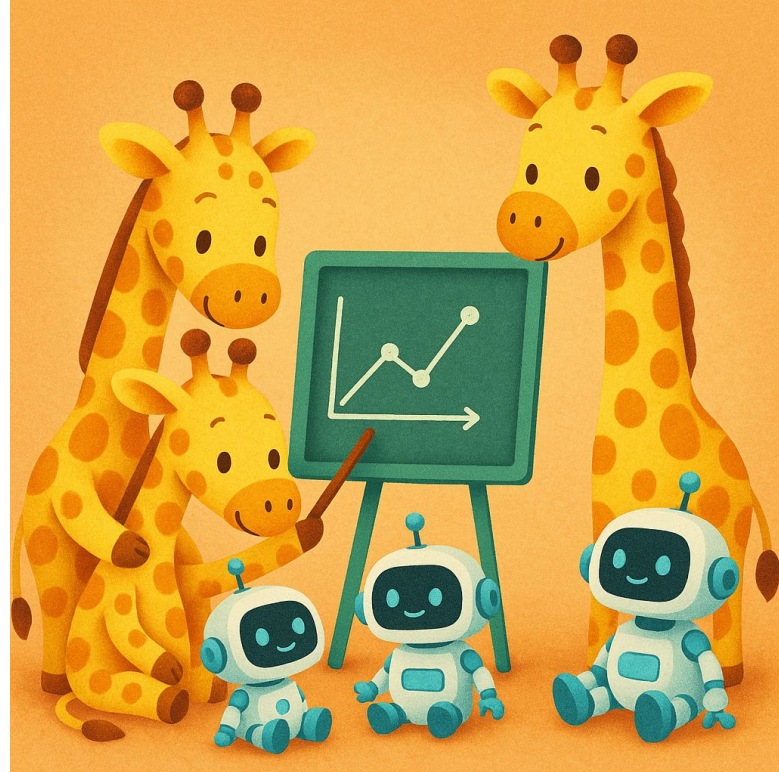
Vladislav Goncharenko

- Founder of girafe-ai
- Author of machine learning courses and Masters program at MIPT
- ML researcher (MIPT)
- Team lead of video ranking team at Dzen (yandex.ru)
- Ex-team lead of perception team at self-driving trucks (Evocargo)
- <https://www.linkedin.com/in/vladislav-goncharenko/>



Rest of the team

- Seminararians
 - give practical sessions
 - Дмитрий Коврижных
- Assistants
 - grade homeworks
- Administrators
 - schedule, paperwork
 - @girafeai_support



How to learn?

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Science based learning techniques

- <https://www.youtube.com/watch?v=ddq8JIMhz7c>
- <https://www.coursera.org/learn/learning-how-to-learn>
- https://wcer.wisc.edu/docs/resources/cesa2017/Dunlosky_SciAmMind.pdf
- https://www.uky.edu/~gmswan3/544/9_ways_to_reduce_CL.pdf

coursera Explore ▾ What do you want to learn? 🔍

🏠 > Browse > Personal Development > Personal Development

DEEP TEACHING SOLUTIONS

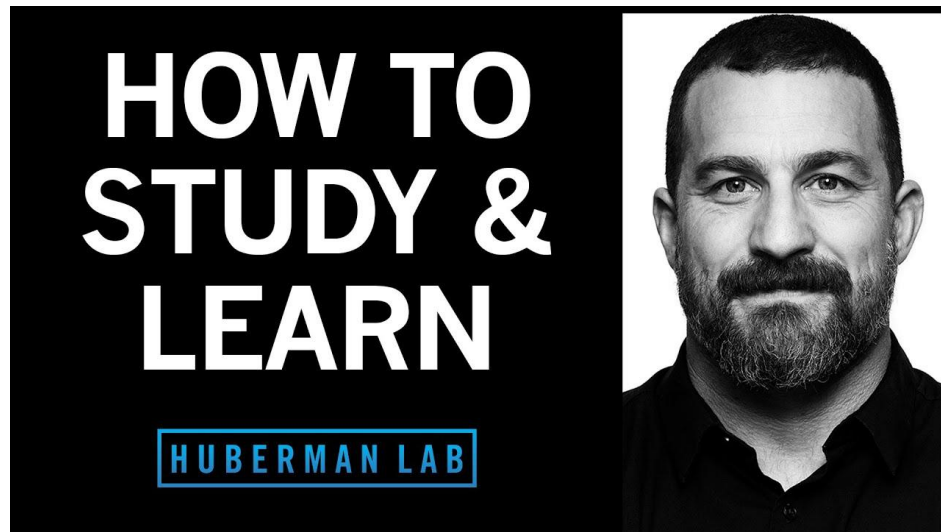
Learning How to Learn: Powerful mental tools to help you master tough subjects

🌐 Taught in English | [22 languages available](#) | Some content may not be translated

Enroll for Free
Starts Sep 13

Financial aid available

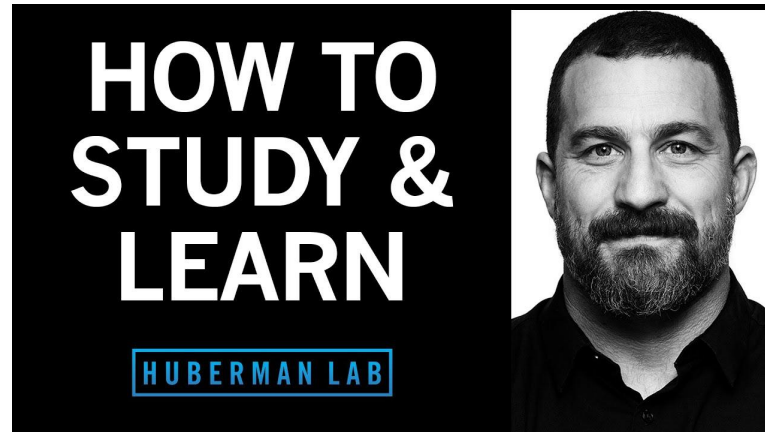
3,849,549 already enrolled



Science based learning techniques

Short outcomes:

- Reserve fixed time to study
- First study alone
- Then try to explain to your classmates
- Answer to questions about the material by yourself



How to work with materials?

It is expected that **you need to study yourself** and no one will do it for you.

Machine Learning sphere is growing explosively and we try to give you the most during this course, so it is full of materials.

It is expected that you will need to **revisit each lecture and practical seminar** by your own. As well as look for some answers in different sources (internet, books, etc...)



Lingua franca

Materials are provided in English because it is the language of international scientific communication thus all the new work is presented in English and we need to stay in line with it.

Also all top scientists in Russia publish all their significant works in English e.g. Ivan Oseledets



Prerequisites

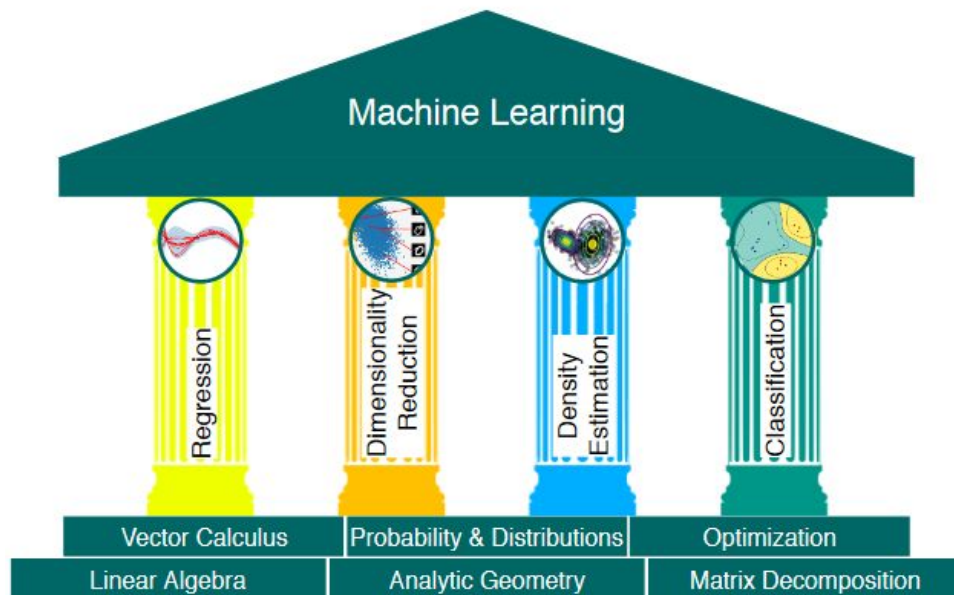
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Math requirements

Basic topics we expect you to know:

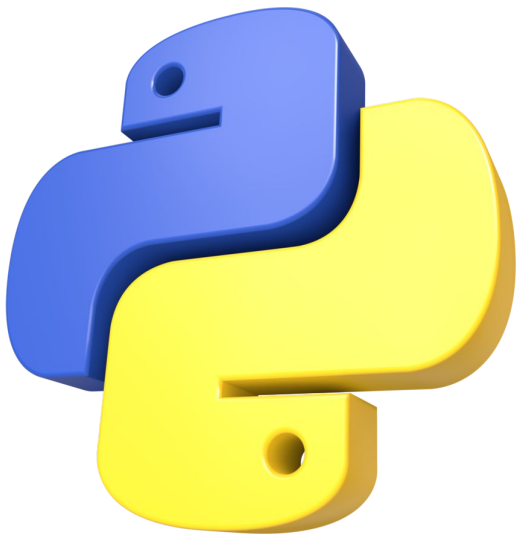
- Linear algebra
 - Matrices
 - Linear functions, transforms
 - Matrix decompositions
- Calculus
 - Matrix differentiation
- Statistics
 - Distributions
 - MLE
- Basics of optimization
 - Extremum
 - Convexity



Materials to refresh necessary math:

- <https://github.com/girafe-ai/math-basics-for-ai/>
- <https://mml-book.github.io/book/mml-book.pdf>

Programming requirements



Tools:

- Python, Numpy, Scipy
- Jupyter notebooks, VS Code
- UNIX terminal, git

Good places to study Python:

<https://snakify.org/en/> or
<https://pythontutor.ru/>

Computer Science basics:

<https://cs50.harvard.edu/x/>

And for UNIX suite:

<https://missing.csail.mit.edu/>

Communication and materials

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Communication structure

Main communication is carried out by Telegram messenger.

1. Course channel
2. Course chat
3. Support account @girafeai_support telegram account

Probably **teacher will not answer to direct messages**
due to large number of students!



Materials

Are published mainly in course repository on github:

https://github.com/girafe-ai/ml-course/tree/25f_mipt

Also some materials can be send to Telegram chat

Videos are published in Youtube channel and Yandex disk.

We have both English and Russian recordings.
If you need them, contact support account.



Course structure

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Course structure

- **Basic track**

- Minimal knowledge of Machine Learning main concepts
- Learning outcome: ability to use existing mainstream ML models
- Duration: 1 term (fall)
- Suitable for programs with ML as minor topic
 - Physics, Chemistry, Biology, etc.

- **Professional track**

- Solid knowledge of Machine Learning
- Learning outcome: ability to research and develop contemporary ML models
- Duration: 2 terms (fall + spring)
- Suitable for programs with ML as major
 - Computer Science, AI, etc.



Basic course structure

1. Introduction, Naive Bayes
2. Linear models
3. Classification and LogReg
4. PCA, kNN, kNN indexes
5. Trees, Bias-Variance-Noise decomposition, inductive bias
6. Bagging, RF, Boosting done right, Contemporary libraries
7. Neural Networks
8. RNN, LSTM, architectures before T. Inductive bias in NNs.
9. Convolution operation. Classification architectures, datasets
10. MLM, Self-attention, Cross-attention mechanisms
11. Ranking
12. Representation learning
13. Unsupervised ML: intrinsic dim, manifold, etc, Geometric DL
14. LLM: decoder, encoder based architectures



Professional course: first term

First term

- Introduction, Naive Bayes
- Linear models
- Classification and LogReg
- PCA and SVM
- **Lect kNN, kNN indexes**
- Trees
- Bias-Variance-Noise decomposition, inductive bias, Bagging, RF
- Boosting done right, Contemporary libraries
- **Hyperparameters tuning. Zero order opt, Optuna**
- Neural Networks
- **Basic text processing & word embeddings as usage of linear layers**
- **Model interpretation, explainable AI (shap, lime and friends; gradCam and friends)**
- Unsupervised ML: intrinsic dim, manifold, etc
- **Advanced instruments: Cython, Numba, Dask**

Professional course: second term

Second term

- Inductive bias in NNs. Optimizers in DL
- RNN, LSTM, architectures before Transformers
- **Classical Image processing and CV. Convolution operation.**
- Classification architectures (ResNet and co) + datasets
- **MLM, Self-attention mechanism**
- **MT, Cross-attention**
- **Geometric DL**
- **Representation learning**
- **Ranking**
- **ML project life cycle and infra (MLOps)**
- **AutoML**
- **Time series**
- LLM: decoder, encoder based architectures
- **Data analytics and EDA**



Other courses and trajectory

- Natural Language Processing, Large Language Models
- Computer Vision
- RecSys and IR
- Timeseries
- MLOps
- Geometric ML
- Digital Signal Processing
- and many more



Grading

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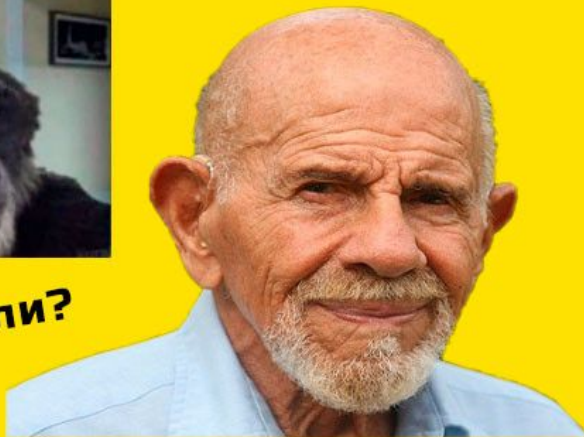
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Загадка от Жака Фреско



че задали?



на размышление даётся 30 секунд

Assignments and grading

Mark	Points	Activities		
		 Tests	 Contest	 Lab
уд	3	60	60	30
хор	5	75	70	50
отл	8	80	80	80

Course structure

Оценка выставляется на экзамене по итогу устного ответа (если это не так, то мы отдельно проговорим это для вас).

Баллы за семестр ограничивают оценку сверху.

“Автоматов” и перезачётов у нас нет.

В силу того, что курс был переработан.

Для самых активных опубликуем форму для написания своей истории.



Экзамен

Устный экзамен по программе. Только на хор и отл.

Программа будет опубликована ближе к концу курса.

На экзамене вам выдадут 3 вопроса из разных разделов программы, на подготовку будет 30 минут, пользоваться можно только своими знаниями.

Далее ответ экзаменатору по подготовленным вопросам, после этого допвопросы (если потребуются).

По итогу экзаменатор выставляет финальную оценку исходя из вашего ответа и баллов за семестр.

Оценка ограничена сверху набранными баллами, баллы сами по себе не могут гарантировать высокую оценку, хотя показывают экзаменатору ваши успехи.



Дипломы с нами

Разные преподаватели имеют разные сферы научных интересов.

Если вы хотите делать проекты (в том числе дипломы) с кем-то - достаточно спросить этого человека.

Также если у вас уже есть тема то мы *возможно* сможем посоветовать кого-то в качестве потенциального научного руководителя.

Постараемся сделать курс для подготовки дипломов.

Список доступных тем опубликуем в канале курса



Telegram channel

<https://bit.ly/ml-course-mipt-25f>



Thanks for attention!

Questions?

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